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Original Research Article

Ceftriaxone-resistant *Salmonella* Typhi isolated from paediatric patients in north India: Insights into genetic profiles and antibiotic resistance mechanismsSushila Dahiya^a, Amit Katiyar^b, Sumit Rai^{c, **}, Priyanka Sharma^d, Punit kaur^d, Arti Kapil^{a, *}^a Department of Microbiology, All India Institute of Medical Sciences, Ansari Nagar, New Delhi, 110029, India^b Bioinformatics Facility, Centralized Core Research Facility, All India Institute of Medical Sciences, Ansari Nagar, New Delhi, 110029, India^c Department of Clinical Microbiology and Infectious Diseases, Super Speciality Child Hospital and Post Graduate Teaching Institute, NOIDA, India^d Department of Biophysics, All India Institute of Medical Sciences, Ansari Nagar, New Delhi, 110029, India

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ABSTRACT

Purpose: To investigate the antibiotic resistance and genetic profile of ceftriaxone-resistant *Salmonella* Typhi isolated from the blood culture of two paediatric cases of typhoid fever and one from the stool culture of their household contact, in North India.

Methods: In this study, whole-genome sequencing was carried out with paired-end 2×150 bp reads on Illumina MiSeq (Illumina, USA) employing v2 and v3 chemistry. To check data quality, adapters and low-quality sequences were removed through Trimmomatic-v0.36. High quality reads were then assembled *de novo* using A5-miseq pipeline. For further refinement, reference-guided contig ordering and orienting were performed on the scaffold assemblies using ABACAS (<http://abacas.sourceforge.net/>). The assembled genome was annotated using Prokka v1.12 to identify and annotate the gene content. Plasmid replicons in bacterial isolates were identified by PlasmidFinder, whereas mobile genetic elements were predicted using Mobile Element Finder. Referenced-based SNP tree with maximum likelihood method was built with CSI phylogeny v1.4.

Results: All three isolates exhibited resistance to ceftriaxone, cefixime, ciprofloxacin, ampicillin, and cotrimoxazole, while demonstrating sensitivity to azithromycin and chloramphenicol. The whole-genome sequencing of these strains revealed the presence of blaCTX-M-15 gene for cephalosporin resistance in addition to gyrA, qnr and IncY plasmid replicon. A 5 kb IS91 *Sbo1* gene cassette (IncY plasmid) was identified which carried extended spectrum β -lactamase blaCTX-M-15, blaTEM-1D (resistance to ampicillin and cephalosporin), sul2, dfrA14 (resistant to trimethoprim-sulfamethoxazole) and qnrS (resistant to ciprofloxacin). These isolates belong to H58 lineage and grouped as sequence type 1 (ST1) on multilocus sequence typing (MLST) analysis.

Conclusion: In the present study we report the isolation of blaCTX-M-15 positive *S. Typhi* from two paediatric patients presenting with fever and one from stool culture of their contact from North India and highlight the need for further investigations to understand the different factors contributing to ceftriaxone resistance in *Salmonella* Typhi.

1. Introduction

Typhoid fever is a systemic febrile illness caused by human restricted *Salmonella* enterica serotype Typhi (hereafter *S. Typhi*). It poses a great public health burden especially on low-income countries of Asia and sub-Saharan Africa, because majority of the population strives for

safe water, hygiene, infrastructure, and limited sanitation [1]. According to a systematic review published in 2019 the pooled incidence of typhoid fever in India was 497.2 (95% CI, 291.9–754.8) typhoid cases per 100,000 per year [2]. Accurate diagnosis followed by appropriate antibiotic treatment is the mainstay of treatment. The chronicle of antibiotic resistance exposes that the introduction of antibiotics induced step-

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wise acquisition of resistance. Initial reports of MDR strains (strain resistant to chloramphenicol, ampicillin, co-trimoxazole) resulted in fluoroquinolones as first line drugs [3,4]. The cost effectiveness of fluoroquinolones made it the first choice of treatment and its usage has increased over the past two decades. This resulted in the emergence of multiple mutants due to selective pressure wherein MDR dominant, H58, emerged dramatically and spread globally. Consequently, the 3rd generation cephalosporins became the main treatment strategy, to which, isolates have remained sensitive till now.

The acquisition of ESBL genes in *S. Typhi* is facilitated by mobile genetic elements such as plasmids. These plasmids can transfer resistance genes between different bacterial strains and species, contributing to the spread of ceftriaxone resistance. The first cephalosporin-resistance cases were reported from Bangladesh in 1998 [5]. However, the recent outbreak of extensively drug-resistant (XDR) typhoid fever in Sindh, Pakistan in 2016 is of concern as travel associated reports of typhoid fever due to the same XDR *S. Typhi* have emerged from different parts of world including UK and USA [6].

Limited reports of ceftriaxone-resistance in *S. Typhi* without genotypic characterization [7,8] and few with genotypic characterization [9–12] have been documented from India. In this study, we report *bla*CTX-M-15 positive *S. Typhi* carrying IncY plasmid replicon from blood culture of two children with enteric fever from the same locality along with a stool culture positive isolate from a household contact of one of them, living in NCR region of India. Further, whole-genome sequencing (WGS) was conducted and phylogenetic relationship of these isolates with other reported XDR *S. Typhi* determined.

2. Methods

2.1. Patient isolates

Ceftriaxone-resistant *S. Typhi* was isolated from blood culture of two children of age 3 years and 5 years from a paediatric superspeciality hospital in North India. They were admitted to a children hospital (NCR India) with enteric fever (hereafter named case-1, and case-2, respectively) following a history of fever for 3 days and abdominal pain. The screening of all family members by stool culture yielded an adult female to be positive (hereafter case-3). The organism identification and antimicrobial susceptibility was recently published by our groups [13]. Here in this study, we present their detailed molecular characterization.

2.2. Whole-genome sequencing (WGS) and surveillance of *S. Typhi*

Whole-genome sequencing was carried out with paired-end 2 × 150 bp reads on Illumina MiSeq (Illumina, USA) employing v2 and v3 chemistry. The generated FASTQ files were quality evaluated by FastQC v0.11.4 [14]. Adapters and low-quality sequences were filtered with Trimmomatic-v0.36 [15]. High quality reads were then assembled *de novo* using A5-miseq pipeline [16]. Reference-guided contig ordering and orienting was done on the scaffold assemblies using ABACAS (<http://abacas.sourceforge.net/>). The quality of the assembled genome was evaluated by QUAST [17] and subsequently annotated using Prokka v1.12 [18]. The genomic predictions of antimicrobial resistance and surveillance of *S. Typhi* were completed through Pathogenwatch [19] available at Center for Genomic Pathogen Surveillance (CGPS, <https://pathogen.watch>). Pathogenwatch provides compatibility with typing information for MLST, cgMLST, *in silico* serotyping (SISTR), SNP genotyping scheme (GenoTyphi), and plasmid replicon sequences to assess public health risk. Plasmid replicons in bacteria isolates were identified by PlasmidFinder, whereas mobile genetic elements and their relation to antimicrobial resistance genes and virulence factors were predicted using MGE [20] available at Center for Genomics Epidemiology (www.genomicsepidemiology.org). Web server oriTfinder (<http://bioinfo-mml.sjtu.edu.cn/oriTfinder/>) was used to identify the origin of transferable antimicrobial resistance in DNA sequences of bacterial mobile genetic elements [21]. Referenced-based SNP tree with maximum likelihood method was built with CSI phylogeny v1.4 [22]. *Salmonella Typhi* CT18 was used as reference genome. The recombination free and referenced-based SNP tree was visualized using FigTree v1.4.3 (<http://tree.bio.ed.ac.uk/software/figtree/>).

3. Results

All three cases were positive for *S. Typhi* on culture. They resided in the same geographical area and had no travel history outside home in recent times. Phenotypically, all three isolates exhibited same antibiotic resistance pattern with resistance to ceftriaxone, cefixime, ciprofloxacin, levofloxacin, ofloxacin, ampicillin and co-trimoxazole and susceptible to azithromycin, and chloramphenicol (Fig. 1a).

3.1. WGS-draft genome and genotypic profiling

The draft genome consists of 4.31 Mb (4309782bp; contigs: 98; GC-content: 52%), 4.30 Mb (4302533bp; contigs: 70; GC-content: 52%) and 4.30 Mb (4302533bp; contigs: 70; GC-content: 52%) genome size for case 1, case 2 and case 3, respectively. All strains belonged to H58 lineage (genotype 4.3.1) and clustered as a monophyletic clade with no genetic diversity. The downstream MLST analysis grouped all the isolates in to sequence type 1 (ST1). AMR profiles and phylogenetic relationship of paediatric isolates with other ceftriaxone resistant isolates are given in (Fig. 1b and c).

3.2. Genomic construction of antibiotic resistance region

The genomic architecture of antimicrobial resistance genes revealed that they belonged to the same group. We found the presence of IncY plasmid replicon in all three isolates carrying ESBL resistance gene on plasmid replicon. A 5 kb IS91 *Sbo1* gene cassette (IncY plasmid) was identified in all cases. All isolates were positive for *bla*CTX-M-15, *bla*TEM-1D (resistant to ampicillin and cephalosporin), *sul2*, *dhfr14* (resistant to trimethoprim-sulfamethoxazole) and *qnrS* (resistant to ciprofloxacin) presented on the gene cassette (Fig. 1d). All isolates carried *gyrA*S83Y mutation and *qnrS1* gene responsible for resistance to ciprofloxacin and other fluoroquinolones (Fig. 1a). The observed genotypic resistance was in accordance with phenotypic resistance.

Additionally, antibiotic resistant genes present in all strains included *mdtK*, *emrR*, *emrB* for fluoroquinolone resistance *kdpE*, APH(3'')-Ib, *baeR*, APH(6)-Id, AAC (6') for aminoglycoside resistance and *tetC* and *mdfA* for tetracycline resistance. The other multiple antibiotic resistant genes obtained include *H-NS*, *sdhA*, *soxS*, *soxR*, *acrA*, *marA*, *marR*, *CRP* and *PBP3* (Supplementary Table 1).

4. Discussion

This study describes ceftriaxone-resistant ESBL producing *S. Typhi* from two children in NCR, India carrying *bla*CTX-M-15 gene and one family member as carrier in gut. These isolates were resistant to ampicillin, cotrimoxazole, ciprofloxacin, levofloxacin, ofloxacin, ceftriaxone and cefixime, but were susceptible to azithromycin, and chloramphenicol in contrast to chloramphenicol-resistance in XDR *S. Typhi*. Hence, the observed pattern did not completely fit to the XDR reported in Pakistan outbreak. This may be a result of increased sensitivity to chloramphenicol in recent years due to the reduced usage of this antibiotic in the Indian subcontinent. In Asia, the overall proportion of chloramphenicol-resistant *S. Typhi* exhibited a declining trend in contrast to a continued rise in MIC (minimum inhibitory concentration) and resistance to fluoroquinolones as ciprofloxacin became the first line drug after the emergence of chloramphenicol resistant *S. Typhi* [23,24]. A

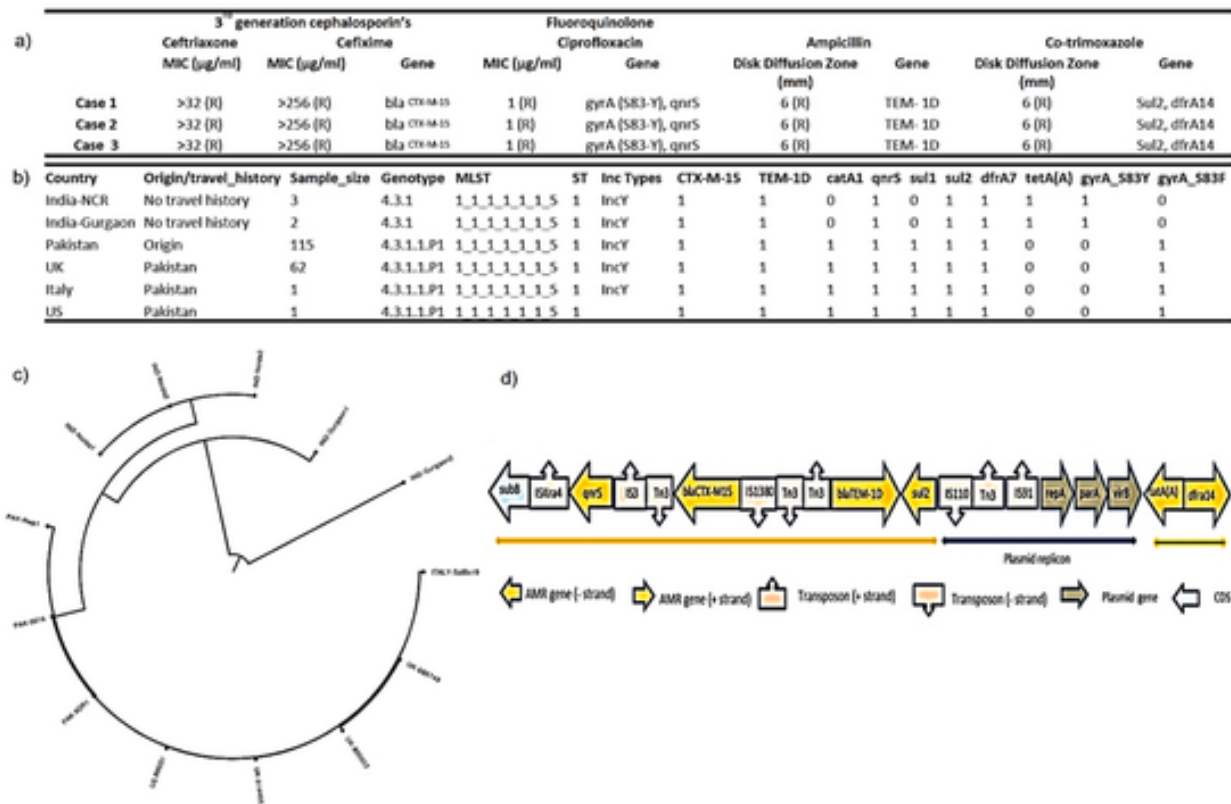


Fig. 1. a) Correlation between antimicrobial susceptibility pattern and AMR genes, b) AMR gene and mutation profile associated with the ESBL-producing *Salmonella* Typhi isolates from India and outbreak isolates of XDR in Pakistan, c) Phylogenetic relationship of studied isolates with other ceftriaxone-resistant isolates reported in literature. All study isolates grouped on same branch with no SNP difference, d) Genetic map of chromosomal antibiotic resistance region with IncY plasmid integrated in study isolates.

study conducted by the ICMR during 2014–2015 at AIIMS, New Delhi, revealed that out of 380 *S. Typhi* strains studied, the overall multiple drug-resistant (resistant to ampicillin, chloramphenicol and co-trimoxazole) *S. Typhi* had decreased to less than 3% [25]. However, with the revision of CLSI breakpoints, the ciprofloxacin susceptibility reduced to 4.6% [23]. The mutations in DNA gyrase contribute the major mechanism responsible for fluoroquinolone resistance with S83Y mutations along with *qnrS1* gene [24]. The increasing ciprofloxacin resistance in *S. Typhi* led to use ceftriaxone as the first line treatment of enteric fever, which has now resulted in increasing MIC to ceftriaxone, creeping towards resistance [23,25]. Recently, sporadic reports of ceftriaxone resistance in *S. Typhi* from India have been reported [7–12]. However, a study from Tamil Nadu, India, has documented *bla*CTX-M-15 in two isolates by PCR but did not report *qnr* gene [10]. Yet in another study, four ceftriaxone resistant *S. Typhi* from Mumbai and one from Vellore were reported to carry IncX3 (*bla*_{SHV-12}, *qnrB7*) and IncN plasmid (*bla*_{TEM-1B}, *bla*_{DHA-1}, *qnrB4*), respectively [11].

In the present study, all three ceftriaxone-resistant strains displayed same resistance profile. We further explored the relatedness between the *bla*CTX-M-15 positive strains reported earlier from India [10], and XDR outbreak strains from Pakistan [5,6] through AMR profiling and SNP analysis. The isolates from this study belong to H58 lineage (genotype 4.3.1) and formed a monophyletic clade, and hence no SNP difference was observed between these strains. The MLST results revealed that the isolates from this clade belonged to sequence type 1 (ST1). In India, ST1 is the most prevalent sequence type of *S. Typhi* followed by ST2 and ST3 [26,27]. The studied strains differed by 32 SNPs and 166 SNPs from Gurgaon01 and Gurgaon02 strains, respectively whilst 31 to 37 chromosomal SNP difference was observed from XDR clade (H58; genotype 4.3.1.1.P1) from Pakistan. The strains isolated from travel associated fever (US, Italy) were also grouped in the same cluster with

XDR clade (H58; 4.3.1.1.P1) and belonged to H58 lineage I. Earlier, ceftriaxone resistant strains from Western and Southern part of India, which do not harbour the *bla*CTX-M-15 gene [8,11] were grouped as ceftriaxone resistant strains in H58 lineage II. This reiterates the fact that H58 lineage II (QRDR triple mutants), is dominant in India. However, isolates from this study and Gurgaon, carrying *bla*CTX-M-15 gene, belonged to genotype 4.3.1. Though studied isolates did not related to XDR isolates but this phylogenetic difference in ceftriaxone resistant *Salmonella* Typhi suggests that the ceftriaxone-resistance in India might have evolved independently and highlights the possibility of multiple genetic mechanisms leading to ceftriaxone-resistance in different lineages of *S. Typhi* circulating within the country. The studied isolates carried the *bla*CTX-M-15 gene on IncY plasmid adjacent to the insertion sequence, ISEcp-1 and ISEc36.

The emergence of ESBL producing *S. Typhi* along with acquired MDR and/or fluoroquinolone-resistance associated determinants and mutations presents a significant treatment challenge. This necessitates antimicrobial susceptibility testing for all *S. Typhi* isolates and treatment of patients accordingly. Azithromycin can be used for the treatment of XDR *S. Typhi* [22] since most isolates in India are currently susceptible to azithromycin except for a few reports of increasing MIC [27,28]. *S. Typhi* with single point mutation R717Q in *acrB* gene responsible for higher MIC against azithromycin (12 µg/ml), reported from Pakistan and Bangladesh, belongs to 4.3.1.1 (H58 lineage I). This represents the same sub lineage as the XDR clade which is currently circulating in Pakistan.

The presence of additional acquired resistance genes for other drugs in *S. Typhi* is of considerable concern, and represents horizontal gene transfer [29]. These findings suggest the evolving threat of new antibiotic resistance in *S. Typhi* and emphasize the increased value of antibiotic susceptibility testing linked to genomic analysis. This makes it im-

perative to develop low-cost diagnostic tools and kits that can be included for rapid detection of such strains. Diagnostic measures in conjunction with the implementation of better public health management approaches like vaccine campaigns along with sanitation and hand hygiene, public health interventions are a prerequisite to bring typhoid under control in endemic regions.

5. Conclusions

In the present study we report *bla*CTX-M-15 positive *S. Typhi* isolated from two paediatric patients presenting with fever and one from stool culture of their contact from North India. A 5 kb IS91 *Sbo1* gene cassette (IncY plasmid) was identified which carried antibiotic resistance gene *bla*CTX-M-15, *bla*TEM-1D (conferring resistance to ampicillin and cephalosporin), *sul2*, *dfrA14* (resistance to trimethoprim-sulfamethoxazole) and *qnrS* (resistance to ciprofloxacin). All isolates belonged to H58 lineage I and grouped as MLST sequence type 1 (ST1). Additionally, all studied isolates showed the presence of the *gyrA*S83Y mutation and the *qnrS1* gene, conferring resistance to fluoroquinolones. Notably, none of the isolates were resistant to azithromycin and chloramphenicol. The presence of IncY plasmid replicon in all three *S. Typhi* isolates carrying ESBL resistance gene and phylogenetic difference within the H58 haplotype suggest that ceftriaxone-resistance in India might have evolved independently and highlights the possibility of multiple genetic mechanisms leading to resistance in different lineages circulating within the country.

At present ceftriaxone along with azithromycin is the main treatment option for treating MDR and fluoroquinolone resistant typhoid fever. The emergence of ESBL producing *Salmonella Typhi* is an alarming situation and need continuous monitoring not only for *Salmonella Typhi* but also of plasmid they carry. Also, understanding the evolutionary pathways leading to ceftriaxone-resistance in this pathogen is crucial for the development of effective treatment strategies. Further studies are needed to elucidate the specific factors contributing to ceftriaxone resistance in different lineages of *Salmonella Typhi* in India, which can help guide strategies for effective management and control of drug-resistant typhoid fever.

Data availability

Whole-genome sequencing data has been submitted to the National Center for Biotechnology Information (NCBI; <https://www.ncbi.nlm.nih.gov/sra>) under BioProject accession number PRJ-NA699425.

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Conflict of interest

All authors declare no conflict of interest.

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Dr. Sumit Rai: Conceptualization, Methodology, Software Sushila Dahiya: Data curation, Writing- Original draft preparation. Dr. Arti Kapil, Dr. Sumit Rai: Visualization, Investigation. Dr. Arti Kapil, Dr. Punit Kaur: Supervision. Sushila dahiya, Amit Katiyar, Priyanka Sharma: Software, Validation. Dr. Arti Kapil, Dr. Punit Kaur: Writing- Reviewing and Editing.

Ethics approval

This study has been approved by the Institutes' ethics committee.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmm.2023.100448>.

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